Reading Plan for Discrete Linear Optimization

Reference

```
@book{bertsimas1997introduction,
  title     = {Introduction to Linear Optimization},
  author    = {Bertsimas, Dimitris and Tsitsiklis, John N.},
  year      = {1997},
  publisher = {Athena Scientific},
  address   = {Belmont, MA},
  series    = {Athena Scientific Series in Optimization and Neural Computation},
  volume    = {6}
}
```

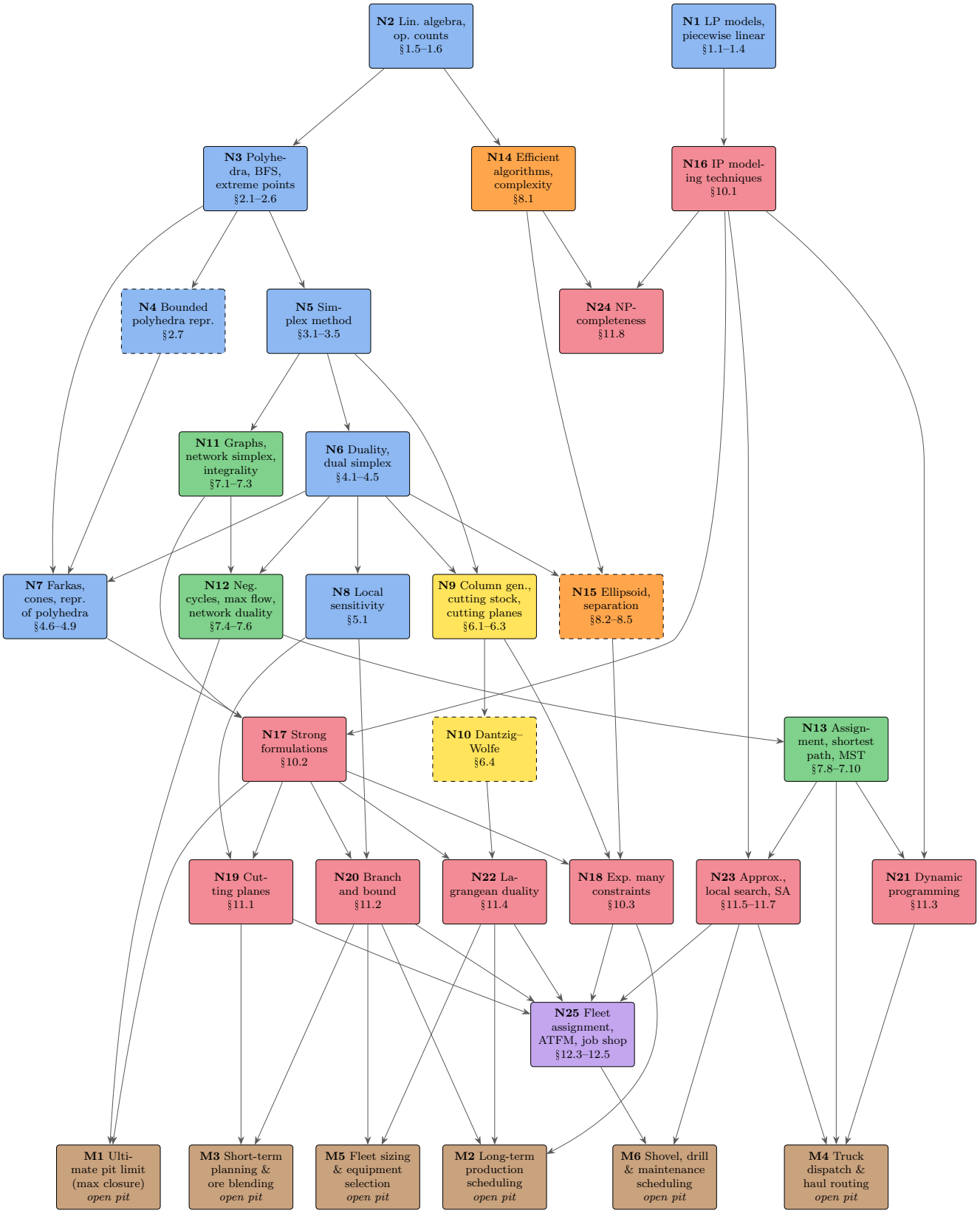
Objective

This plan selects the parts of [1] that build the toolkit of discrete linear optimization. After following it you should be able to use:

- **Polyhedral theory:** extreme points and basic feasible solutions, representation of polyhedra, Farkas' lemma.
- **LP algorithms and duality:** simplex and dual simplex, duality bounds, reoptimization after adding a variable or a constraint.
- **Large-scale LP:** column generation, constraint generation, Dantzig–Wolfe decomposition.
- **Network optimization:** network simplex, max flow/min cut, assignment, shortest paths, minimum spanning trees, and the integrality of network flows.
- **Integer modeling:** binary-variable formulations and comparison of formulations by the strength of their LP relaxations.
- **Exact integer methods:** cutting planes, branch and bound, dynamic programming, Lagrangean relaxation.
- **Heuristics:** approximation algorithms, local search, simulated annealing.
- **Complexity:** polynomial-time algorithms, separation and the ellipsoid method, NP-completeness, to judge which method a problem admits.

Not covered: Chapter 9, §2.8, §3.6–3.7, §5.2–5.5, §6.5, §7.7, §12.1–12.2 (1997 edition numbering).

1 Reading paths



Legend. LP foundations Networks Large scale Complexity Integer programming Book applications Open-pit mining (outside the book). Dashed box: optional, can be dropped if time is short.

2 Node coverage table

Hours: estimated time to read the sections and solve a few exercises per node. Total: 191 hours.

Node	Sections	Content covered	Role in discrete optimization	Hours	Status
N1	1.1–1.4	LP forms, modeling examples, piecewise linear convex objectives, graphical solution	Modeling vocabulary reused for IP	4	Core
N2	1.5–1.6	Notation, linear algebra, operation counts	Basis for complexity arguments	4	Support
N3	2.1–2.6	Polyhedra, extreme points = vertices = BFS, degeneracy, optimality of extreme points	Integrality is a question about extreme points	12	Core
N14	8.1	Input size, polynomial vs. exponential algorithms	Prerequisite for NP-completeness	3	Core
N16	10.1	Binary choice, forcing constraints, disjunctions, piecewise linear costs	Turning logic into linear constraints	5	Core
N4	2.7	Bounded polyhedron = convex hull of extreme points	Underlies conv(S)	3	Optional
N5	3.1–3.5	Reduced costs, simplex, tableau, anticycling, Phase I	Tableau used by Gomory cuts and network simplex	16	Core
N24	11.8	P, NP, NP-completeness, reductions	Explains why IP is hard	8	Core
N6	4.1–4.5	Duality, complementary slackness, shadow prices, dual simplex	Bounds, reoptimization, network duality	16	Core
N11	7.1–7.3	Graphs, network flow formulation, network simplex, integral flows	Integer problems solvable as LPs	10	Core
N7	4.6–4.9	Farkas' lemma, cones, extreme rays, resolution theorem	Polyhedral description of conv(S)	10	Core
N8	5.1	Adding a variable/constraint, changes in b, c	The operations of cutting and branching	5	Core
N9	6.1–6.3	Column generation, cutting stock, constraint generation	Large formulations; prototype of separation	8	Core
N12	7.4–7.6	Negative cost cycles, max flow, min cut, network duality	Min–max theorems via LP duality	8	Core
N15	8.2–8.5	Ellipsoid method, separation, exponentially many constraints	Separation $\equiv$ optimization	6	Optional
N10	6.4	Dantzig–Wolfe decomposition	Linked to Lagrangean duality	5	Optional
N13	7.8–7.10	Assignment, auction, shortest path, MST	Combinatorial subproblems	8	Core
N17	10.2	Comparing formulations; ideal formulation = conv(S)	Central concept of the plan	12	Core
N18	10.3	Formulations with exponentially many constraints	Needs separation/cutting planes	6	Core
N19	11.1	Gomory cutting planes	Exact method from tableau rows	6	Core
N20	11.2	Branch and bound	Workhorse algorithm for IP	10	Core
N21	11.3	Dynamic programming (knapsack)	Pseudo-polynomial exact method	4	Core
N22	11.4	Lagrangean relaxation and dual	Stronger bounds than LP	10	Core
N23	11.5–11.7	Approximation, local search, simulated annealing	Methods for large instances	6	Support
N25	12.3–12.5	Fleet assignment, air traffic flow, job shop	Formulation + methods together	6	Support

3 Open-pit mining problems

The bottom nodes of the graph (M1–M6) are open-pit mining problems that the techniques above can solve. For each one: the path in the graph that leads to it, what the problem asks, and why it needs the nodes with a direct arrow into it.

Ultimate pit limit (M1).
Path in the graph: N2 → N3 → N5 → N11 → N12 → M1, with N7 → N17 → M1.

Objective: the most valuable set of blocks that respects slope precedence.
Why the parents: the problem is a maximum closure, which reduces to a minimum cut [3]; N12 gives max-flow/min-cut. N17 explains why the precedence formulation is already ideal, so the LP relaxation returns an integer pit without branching.

Long-term production scheduling (M2).
Path in the graph: N6 → N9 → N10 → N22 → M2, with N17 → N18 → M2 and N17 → N20 → M2.
Objective: assign blocks to periods to maximize NPV under precedence and capacity.
Why the parents: N18 handles the huge number of precedence constraints; N22 relaxes capacities so each subproblem becomes a pit (closure) problem, which gives bounds [2]; N20 closes the remaining integrality gap.

Short-term planning and ore blending (M3).
Path in the graph: N5 → N6 → N8 → N19, N20 → M3, with N17 → N19, N20.
Objective: choose faces or polygons per week or shift that meet tonnage and grade bounds.
Why the parents: N19 strengthens the relaxation with cuts; N20 fixes the binary extraction decisions; both rely on N8 to reoptimize after adding a cut or a branch.

Truck dispatching and haul routing (M4).
Path in the graph: N11 → N12 → N13 → M4, with N13 → N21, N23 → M4.
Objective: assign trucks to shovels and dumps, and pick haul routes, in real time.
Why the parents: N13 gives shortest paths (routes) and assignment (truck–shovel); N21 models sequential decisions, the bridge to stochastic DP and RL; N23 gives fast heuristics when exact solutions are too slow.

Fleet sizing and equipment selection (M5).
Path in the graph: N17 → N20 → M5, with N9 → N10 → N22 → M5.
Objective: integer numbers of trucks and shovels per period at minimum cost.
Why the parents: N20 handles the integer fleet counts; N22 decomposes the problem by period and gives lower bounds.

Shovel, drill and maintenance scheduling (M6).
Path in the graph: N20, N22, N18, N19, N23 → N25 → M6, with N23 → M6.
Objective: sequence equipment moves and maintenance windows.
Why the parents: N25 (§12.5, job shop) gives the scheduling formulation; such problems are NP-hard, so N23 supplies the heuristics used in practice.

References

[1] Dimitris Bertsimas and John N. Tsitsiklis. *Introduction to Linear Optimization*, volume 6 of *Athena Scientific Series in Optimization and Neural Computation*. Athena Scientific, Belmont, MA, 1997.

[2] Daniel Bienstock and Mark Zuckerberg. Solving LP relaxations of large-scale precedence constrained problems. In *Integer Programming and Combinatorial Optimization (IPCO 2010)*, volume 6080 of *Lecture Notes in Computer Science*, pages 1–14. Springer, 2010.

[3] Jean-Claude Picard. Maximal closure of a graph and applications to combinatorial problems. *Management Science*, 22(11):1268–1272, 1976.